

SIMATS ENGINEERING

CO3- AT2 - Design Task - Medium

COURSE NAME:5G / 6G Communication Systems

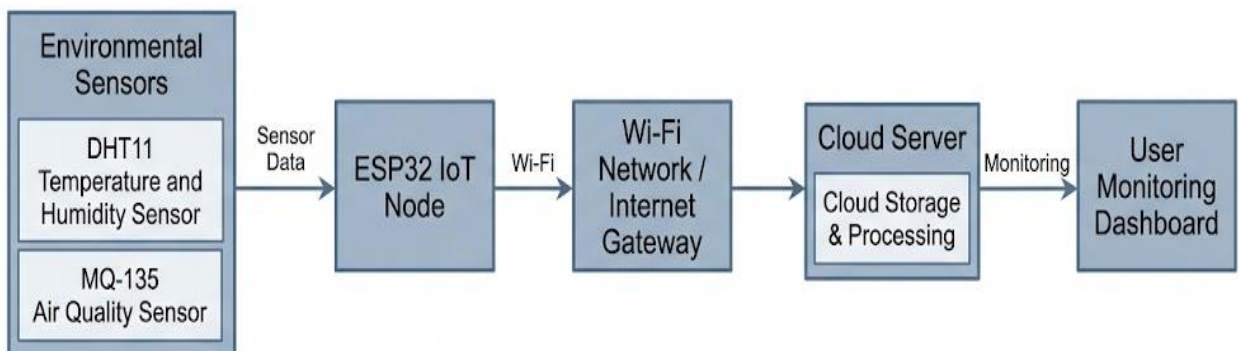
COURSE CODE: ECA5607

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31. Design an IoT communication flow for periodic environmental sensing. (BL6)

Design and Implementation of an IoT Communication Flow for Periodic Environmental Sensing



MATLAB CODE:

```
clc;

clear;

close all;

% IoT Periodic Environmental Sensing Simulation

N = 20;

T = 10;

time = (0:N-1) * T;

temperature = 25 + 2 * randn(1,N);

humidity = 60 + 5 * randn(1,N);

airQuality = 300 + 30 * randn(1,N);

temperature = max(min(temperature,40),15);

humidity = max(min(humidity,90),30);

airQuality = max(airQuality,100);

fprintf('IoT Environmental Monitoring Simulation\n');

fprintf('-----\n');

fprintf('Time(s)\tTemperature(C)\tHumidity(%%)\tAir Quality\n');

for i = 1:N

    fprintf('%d\t%.2f\t%.2f\t%.2f\n', ...

        time(i), temperature(i), humidity(i), airQuality(i));

end

figure;

plot(time, temperature, '-o');

grid on;

xlabel('Time (seconds)');

ylabel('Temperature (C)');

title('Periodic Temperature Monitoring');

figure;

plot(time, humidity, '-o');
```

```
grid on;
xlabel('Time (seconds)');
ylabel('Humidity (%)');
title('Periodic Humidity Monitoring');
figure;
plot(time, airQuality, '-o');
grid on;
xlabel('Time (seconds)');
ylabel('Air Quality');
title('Periodic Air Quality Monitoring');
figure;
plot(time, temperature, '-o');
hold on;
plot(time, humidity, '-s');
plot(time, airQuality, '-^');
grid on;
xlabel('Time (seconds)');
ylabel('Sensor Values');
title('IoT Periodic Environmental Sensing');
legend('Temperature', 'Humidity', 'Air Quality');
```

SIMULATION OUTPUT:

